1. A cart of mass 2 kg moves at 3 m/s on a level track. Find its kinetic energy.

2. The cart then climbs a ramp. How high does it go before it stops?

3. Repeat question 2 for a cart of twice the mass and explain the result.

Displacement is the change in position of an object. It has a direction as well as a size, so it is a vector. Distance is how far the object travels along its path. Average velocity is displacement divided by the time it takes. Speed is the size of the velocity. A car that drives 30 m east in 3 s has an average velocity of 10 m/s.

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